

PARTHO SARKER DHRUBO

Bioinformatics Researcher & Agricultural Innovator | PSTU Research Society President

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PROFILE

Motivated bioinformatics researcher with hands-on experience in molecular docking, phylogenetic analysis, and agricultural biotechnology. President of the PSTU Research Society with proven leadership in organizing academic conferences and industry collaborations. Passionate about bridging computational biology and sustainable agriculture through data-driven research and technology.

EDUCATION

Patuakhali Science and Technology University (PSTU) — Undergraduate Program

Relevant Coursework: Molecular Biology, Bioinformatics, Plant Biotechnology, Biostatistics, Research Methodology

RESEARCH EXPERIENCE

Molecular Docking Analysis of *Fusarium graminearum*

- Performed in-silico molecular docking to identify potential inhibitory compounds against *Fusarium graminearum*, a devastating fungal pathogen affecting cereal crops.
- Utilized AutoDock Vina and PyMOL for binding affinity scoring and 3D visualization of ligand-receptor interactions.
- Analyzed binding energies and identified lead compounds with high inhibition potential for further validation.

11th International Plant Tissue Culture & Biotechnology Conference — University of Dhaka

- Presented a research poster on cutting-edge plant biotechnology techniques and tissue culture applications.
- Received certification for active participation and scholarly contribution to the conference proceedings.

Chili Quality & Biochemical Profiling Research

- Conducted TSS (Total Soluble Solids) estimation and ascorbic acid profiling in chili varieties.
- Executed AAS (Atomic Absorption Spectroscopy) operations for mineral content analysis.
- Designed and managed pot experiments with optimized treatment setups and rigorous trait analysis protocols.

PROJECTS

Krishibondhu (■■■■■■■■■■) App — Agricultural Tech Concept

- Conceptualized a farmer-centric mobile application to deliver real-time agricultural advisory, crop monitoring, and market linkage services to rural farming communities in Bangladesh.
- Focused on bridging the information gap between agricultural researchers and field-level farmers using accessible technology.

LEADERSHIP & EXTRACURRICULAR

President, PSTU Research Society

- Led university-wide research initiatives, mentored junior researchers, and coordinated interdisciplinary academic projects.
- Organized workshops, seminars, and collaborative events to foster a culture of scientific inquiry.

Journey to Career — Collaboration with bdjobs (2025)

- Coordinated the 'Journey to Career' program in partnership with Bangladesh's largest job-seeking platform, bdjobs.com.
- Facilitated career development sessions connecting students with industry professionals and employment opportunities.

Qamarun Agro Farm — 4th Youth Business Conference (2025)

- Participated in youth entrepreneurship programming, gaining exposure to agribusiness operations and sustainable farming enterprises.

TECHNICAL SKILLS

Bioinformatics Tools: BLASTp, MEGA12 (phylogenetics), trRosetta (protein structure prediction), AutoDock Vina, PyMOL, molecular docking, sequence alignment, phylogenetic tree construction

Laboratory Skills: Fluorescence-based chlorophyll content determination, AAS operation, UV/VIS spectrophotometry, TSS estimation, ascorbic acid profiling, dose optimization, pot experiments, research design, sample analysis, treatment setup, trait analysis

Research & Analysis: Experimental design, statistical data analysis, biochemical profiling, scientific writing, poster presentation

Soft Skills: Team leadership, public speaking, academic event organization, cross-functional collaboration

CERTIFICATIONS & CONFERENCES

- Certification — 11th International Plant Tissue Culture & Biotechnology Conference, University of Dhaka
- Poster Presentation — International Plant Tissue Culture & Biotechnology Conference
- Participation — 4th Youth Business Conference, Qamarun Agro Farm (2025)
- Collaboration — Journey to Career Program with bdjobs.com (2025)

LANGUAGES

Bengali (Native) — English (Professional Working Proficiency)

References available upon request.